SYSTEM SPECIFICATION

ZCOS Memory Engine

Canonical implementation requirements for object-based, galaxy-partitioned memory

Field	Specification value
Status	Locked for implementation
Authority	ZCOS Architecture Foundation
Scope	Memory capture, governance, partitioning, lifecycle, retrieval, correction, and forgetting
Version date	August 7, 2026
Migration source	Existing ZAR Object Memory and related memory services

Governing rule: One ZCOS Memory Engine. Eight isolated galaxy partitions. One canonical object record per retained memory. Controlled cross-galaxy access. Complete provenance. Cascading correction and deletion.

1. Purpose and Authority

This specification defines the required behavior and data contract for the ZCOS Memory Engine. It translates the locked ZCOS Architecture Foundation into an implementation-ready system contract. It governs what qualifies as Memory, how Memory is represented, how it enters and leaves the system, and how every read or write is authorized.

The ZCOS Architecture Foundation is authoritative when this specification conflicts with older ZAR documents or repository behavior. The existing ZAR specification and repository are migration evidence: their reusable object, source, relationship, isolation, and retrieval mechanisms should be preserved only where they comply with this contract.

1.1 Required outcomes

- Preserve the strongest parts of the existing Object Memory implementation without retaining its mixed-domain authority.
- Give each authenticated user one logically unified Memory system divided into eight galaxy partitions.
- Keep the user-facing experience simple: You, Topics, Galaxies, Memory Summary, and direct remember commands.
- Make every retained memory traceable, correctable, supersedable, reviewable, and forgettable.

- Prevent disabled, unauthorized, rejected, superseded, or forgotten records from influencing responses.

2. Non-Negotiable Invariants

Invariant	Required meaning
One engine	ZCOS owns one Memory Engine. No galaxy, workspace, feature, file path, or vector database becomes a competing memory authority.
Authenticated ownership	Every user-owned memory operation requires a verified owner user ID. Missing or fallback owners must fail closed.
Galaxy partitioning	Every memory belongs to exactly one originating galaxy partition at creation. Cross-galaxy access never changes origin.
Canonical object record	The authoritative memory is a durable object record. Search indexes, summaries, prompts, caches, and embeddings are derived representations.
Provenance required	No canonical memory may exist without at least one source ledger entry or an explicit user-directed source.
State-safe retrieval	Only records allowed by lifecycle, toggle, ownership, partition authorization, and retention policy may be retrieved.
No silent overwrite	Corrections and conflicts preserve history. Newer information may supersede an older record but may not erase the record's existence without a forget action.
Deletion propagation	Forgetting removes the canonical content from active use and invalidates every derived index, cache, summary, and unauthorized replica.
Domain separation	Identity, Knowledge, Glow, Settings, Projects, Files, and conversation history remain separate authorities.

3. Scope and Domain Boundaries

3.1 Canonical Memory types

Type	What it retains	Examples of qualifying content
Experience	Something the user experienced or reports having experienced.	A meaningful personal or work experience; a recurring situation.
Decision	A choice that was made, including its context and rationale when available.	A selected product direction; a rejected option; a governing decision.
Person / Relationship	A person and the user's relationship or interaction context.	A collaborator, family relationship, professional contact, or recurring interpersonal context.

Type	What it retains	Examples of qualifying content
Event	Something that occurred and may matter later.	A deployment, meeting outcome, milestone, failure, move, or other dated occurrence.
User-directed	Anything the user explicitly instructs ZCOS to remember.	A direct 'remember that...' instruction, subject to safety and domain routing.

3.2 Items routed elsewhere

Content	Canonical owner
Name, email address, profile image, Privy wallet address	Identity
Facts, claims, concepts, systems, rules, sources, substantiated relationships	Knowledge
Explicit interaction preferences and operating controls	Settings
Inferred personal patterns, adaptation, and growth	ZWAP! Discovery -> Glow
Tasks, open questions, project state, work products	Desk / Projects
Original files and stored artifacts	ZENITH Logos -> Scholar -> Files
Conversation transcripts and message history	Conversation History; usable as a source, not automatic Memory
System identity, policies, reasoning rules, tool contracts	ZCOS Core / system governance, not user Memory

Boundary rule: A source may produce more than one routed object, but each resulting object must have one canonical domain owner. Linking does not merge authorities.

4. Engine Architecture

4.1 Required layers

Layer	Responsibility	Authority status
Capture and classification	Receives eligible signals, determines whether a memory candidate exists, classifies type and domain, and assigns a galaxy.	Processing only

Layer	Responsibility	Authority status
Source ledger	Records where the memory came from and the exact source linkage.	Authoritative provenance
Canonical memory store	Stores the durable object record, lifecycle, ownership, partition, versions, and links.	Canonical Memory authority
Relationship graph	Connects memories to memories and to authorized external-domain entities.	Canonical relationships; not a second store
Retrieval indexes	Supports semantic, keyword, topic, entity, date, and relationship lookup.	Derived and rebuildable
Audit ledger	Records state transitions, access, cross-galaxy authorization use, correction, export, and deletion actions.	Authoritative audit evidence
User surfaces	Presents You, Topics, Galaxies, Summary, review, direct input, correction, and forgetting.	View and control layer

4.2 Processing sequence

- Accept an eligible source event from an authenticated context.
- Check the global Memory toggle before any long-term extraction or retrieval.
- Classify the signal as temporary context, conversation history, Memory, or another canonical domain.
- If Memory is eligible, assign owner, originating galaxy, type, provenance, topics, entities, and initial lifecycle state.
- Compare the candidate with existing canonical records for duplicates, conflicts, corrections, and supersession.
- Persist the source ledger and canonical record atomically, then update derived indexes.
- Retrieve only through the authorization and lifecycle gates defined in this specification.

5. Ownership and Galaxy Partitions

5.1 Partition set

Partition	Default access
ZAR	ZAR Nexys reads and writes ZAR Memory.
ZYNC	ZYNC Canvas reads and writes ZYNC Memory.
ZETA	ZETA Control reads and writes ZETA Memory.
ZENO	ZENO Unite reads and writes ZENO Memory.

Partition	Default access
ZYLO	ZYLO Compass reads and writes ZYLO Memory.
ZWAP!	ZWAP! Discovery reads and writes ZWAP! Memory.
ZENITH	ZENITH Logos reads and writes ZENITH Memory.
ZILLION	ZILLION Prosper reads and writes ZILLION Memory.

5.2 Partition rules

- The originating galaxy is assigned from the authenticated execution context, not inferred from keywords when the active galaxy is known.
- A record may reference entities or projects owned elsewhere without moving into that other partition.
- A cross-galaxy read, write, or contribution requires an active authorization from ZCOS Command Desk -> Admin Access.
- An authorization grants access; it does not duplicate or relabel the canonical record.
- All Memory may query every partition for the authorized administrator while preserving each result's origin and access history.
- Authorization revocation must stop future access immediately and invalidate derived cross-galaxy caches or summaries.

5.3 Fail-closed ownership

User-owned reads and writes must reject absent, malformed, invented, or known fallback identities. Existing patterns such as user, user_001, default-user, anonymous, unknown, offline, and admin-user are not valid substitutes for authentication. External channels must resolve to a verified unified Identity before they may access long-term Memory.

6. Canonical Memory Record Contract

6.1 Required fields

Field	Requirement
id	Stable, unique object identifier.
ownerUserId	Verified unified Identity owner.
galaxyId	Exactly one originating galaxy partition.
memoryType	Experience, decision, person_relationship, event, or user_directed.
canonicalName	Short stable label suitable for browsing and linking.
content	Canonical retained meaning in clear language.

Field	Requirement
properties	Type-specific structured values; never the sole location of required core fields.
topics	Zero or more topic identifiers used to organize the record.
entityIds	Links to people, projects, or other authorized canonical entities.
sourceRefs	One or more source ledger references.
confidence	Evidence-proportional extraction confidence; not a substitute for lifecycle state.
occurredAt	When the remembered event or condition occurred, if known.
createdAt / updatedAt	System timestamps for record creation and latest version.
lifecycleState	Proposed, active, confirmed, corrected, superseded, rejected, or forgotten.
confirmationMethod	Direct user instruction, user review, policy-approved automatic retention, correction, or migration.
supersedesId	Record replaced by this version, when applicable.
derivedFromIds	Prior records used to form this record, when applicable.
retentionPolicy	Applicable retention behavior and review conditions.
deletedAt / deletionCascadeId	Forgetting metadata when the state is forgotten.
version	Monotonic record version or equivalent immutable revision reference.

6.2 Type-specific minimums

- Decision: the decision itself; decision date when known; rationale when known; affected entities when known.
- Person / Relationship: referenced person identity or alias; relationship context; source boundaries that prevent unsupported identity merging.
- Event: event description; occurredAt when known; result or consequence when available.
- Experience: concise experience statement; time range when known; relevant people, topics, and context.
- User-directed: the remembered instruction and the exact direct-command source reference.

7. Relationships and Object Graph

The existing object-oriented graph remains reusable infrastructure. Relationships must be first-class records with their own IDs, endpoints, type, source or evidence, confidence, creation time, and lifecycle validity.

7.1 Required relationship semantics

Relationship	Meaning
RELATED_TO	General meaningful association when no stronger defined relation applies.
INVOLVES	A memory involves a person, project, entity, or topic.
DERIVED_FROM	A record was compiled from another record or source-backed memory.
SUPERSEDES	A newer record replaces an older record's active meaning.
CONTRADICTS	Two records cannot both be accepted without review or context separation.
AFFECTS	A decision or event changes another remembered entity or situation.
OCCURRED_IN	An experience or event occurred within a project, conversation, location, or time context.

Existing relationship types may remain in the shared framework for Knowledge or Projects, but Memory retrieval must interpret only relationships valid for the canonical domains involved. A relationship cannot bypass partition authorization.

8. Memory Lifecycle

State	Entry condition	Retrieval behavior
Proposed	ZCOS inferred a candidate that requires review or additional evidence.	Not used as user truth; visible in review where applicable.
Active	Retained automatically under the user's approved Memory policy.	Eligible when all authorization and relevance gates pass.
Confirmed	The user explicitly directed or confirmed the memory.	Eligible and favored over unconfirmed conflicting records.
Corrected	The user or an authorized process corrected the retained meaning.	Corrected version is eligible; prior version remains historical.
Superseded	A newer record replaced the current meaning.	Excluded from ordinary retrieval; available for history and explanation.
Rejected	The candidate was determined not to be retained Memory.	Excluded from retrieval and future automatic promotion from the same evidence unless new evidence changes classification.
Forgotten	The user deleted the memory and the deletion cascade was initiated.	Never retrievable as content; only minimal audit evidence may remain where required.

8.1 Transition rules

- Direct 'remember' input enters Confirmed after authentication, domain routing, and safety validation.
- Automatic retention may enter Active only when Memory is enabled and the configured policy allows it.
- Low-confidence, conflicted, or materially ambiguous candidates enter Proposed, not Active.
- Correction creates a new authoritative revision and preserves the prior revision as historical evidence.
- Supersession requires an explicit link from the new active record to the replaced record.
- Rejection and forgetting are distinct: rejection prevents candidate promotion; forgetting removes a previously retained memory from active and derived use.

9. Capture, Classification, and Consolidation

9.1 Eligible sources

- Direct user memory commands.
- Authenticated conversations while Memory is enabled.
- Authorized galaxy actions and system events.
- Files or imports only after source ownership, domain routing, and ingestion approval are established.
- Legacy archives only through an explicit migration process; never as automatic current truth.

9.2 Candidate tests

Before creating a canonical memory record, the engine must determine whether the information is durable enough to matter later, belongs to Memory rather than another domain, has an identifiable owner and galaxy, and is supported by traceable source evidence. Temporary instructions, speculative statements, quoted third-party claims, and ordinary conversational filler must not be promoted as personal truth.

9.3 Duplicate and conflict handling

- Exact or near-duplicate candidates append provenance to the existing record when their meaning is materially the same.
- A changed value becomes a correction or supersession when the user or reliable temporal context establishes a new state.
- Incompatible claims become a conflict requiring context separation or review; neither is silently overwritten.
- Confidence increases only when independent evidence supports the same meaning. Repetition from one copied source is not independent corroboration.
- The engine must track state over time instead of flattening historical and current truth into one value.

10. Source Ledger and Provenance

10.1 Source reference requirements

Source field	Required behavior
sourceType	Conversation, action, file, system_event, direct_user, or migration.
sourceId	Stable identifier for the source artifact or event.
sourceOwnerUserId	Owner of the source when user-owned.
sourceGalaxyId	Galaxy context in which the source entered ZCOS.
location	Message, page, segment, event position, or equivalent locator when available.
evidenceExcerpt	Minimal exact excerpt or normalized evidence required to verify extraction; governed by privacy and retention rules.
capturedAt	When the engine recorded the source reference.
occurredAt	When the source-described event occurred, if known.
extractionMethod	Direct instruction, deterministic rule, model extraction, migration, or manual confirmation.

10.2 Provenance rules

- A memory may have several source references, but a source reference may not silently become ownership evidence for another user.
- Deleting a conversation does not automatically delete a separately confirmed memory unless the user's data-control choice requires that cascade.
- Forgetting a memory invalidates derived excerpts and indexes even if its original source remains under separate retention controls.
- Exports must preserve origin, lifecycle, and source references sufficiently to reconstruct meaning and governance.

11. Retrieval Contract

11.1 Mandatory retrieval gates

- Verified user ownership.
- Memory enabled for the user.
- Originating partition matches the active galaxy or an active cross-galaxy authorization exists.
- Lifecycle state is eligible for the requested operation.

- Retention policy permits access.
- The record is relevant to the current objective, not merely keyword-adjacent.
- No newer confirmed or corrected record supersedes it.
- The response context receives only the minimum relevant memory content and provenance needed.

11.2 Ranking requirements

Retrieval may combine semantic, keyword, topic, entity, relationship, and temporal indexes. Ranking should consider relevance, lifecycle authority, confirmation, confidence, recency where relevant, and relationship context. Confidence alone must not outrank a directly confirmed correction. Keyword overlap alone must not establish relevance.

11.3 Response use

- Retrieved Memory is context, not an instruction to ignore current user input or higher governing policy.
- When retrieved memories conflict and the difference could change the answer, ZCOS must ask a concise question or explain the uncertainty.
- The system must not expose internal graph IDs, embedding scores, storage paths, or hidden reasoning unless a legitimate administrative surface requires them.
- Every response trace must be able to identify which memory IDs were retrieved without exposing that trace by default.

11.4 Index requirements

- Indexes must be reproducible from the canonical store and relationship graph.
- Vector embeddings are replaceable derived indexes, never canonical Memory.
- Index updates must occur after successful canonical writes and must be retried or reconciled when partial failure occurs.
- Superseded, rejected, and forgotten records must be excluded or explicitly filtered from ordinary search results.

12. Memory Toggle and Settings Contract

Setting or action	Required behavior
Enable Memory: On	Eligible sources may create long-term memories and authorized records may be retrieved.
Enable Memory: Off	No new long-term extraction and no retrieval of existing memory during responses.
Memory Summary	Plain-language overview derived from eligible canonical records; never a separate memory store.
Manage Memory	Opens the Memory domain organized as You, Topics, and Galaxies.

Setting or action	Required behavior
Tell ZAR what to remember	Creates a Confirmed user-directed memory in the active galaxy after validation and routing.
Re-enable Memory	Restores authorized access to existing retained memory; does not backfill conversations created while disabled.

Turning Memory off does not delete existing records. Conversation history has separate retention controls and must not be treated as long-term Memory solely because it remains stored.

13. User-Facing Memory Domain

You, Topics, and Galaxies are three projections over the same canonical records. They must never create duplicate records or separate stores.

Area	Required contents
You	Personal experiences, decisions, people and relationships, and user-directed memories.
Topics	Memories grouped by recurring subjects, current work, projects, and ongoing situations.
Galaxies	Entry to each of the eight actual galaxy partitions, with origin visibly preserved.

13.1 Required controls

- View the concise canonical memory and its originating galaxy.
- View understandable source information and relevant dates without exposing internal storage details.
- Correct a memory.
- Confirm or reject a proposed memory when review is required.
- Forget a memory with clear scope and downstream consequences.
- Navigate related memories, topics, people, projects, and source history when authorized.

14. Correction, Supersession, and Forgetting

14.1 Correction

- Create a new revision rather than mutating historical evidence beyond recognition.
- Mark the corrected revision as authoritative and record who or what initiated the correction.
- Update affected summaries, relationships, and indexes.
- Prevent the prior meaning from being used as current truth while allowing authorized historical explanation.

14.2 Forgetting

- Authenticate the owner and resolve the exact target before deletion.
- Set the record to Forgotten and begin a durable deletion cascade.
- Remove canonical content from active storage according to the deletion policy while retaining only the minimum non-content audit marker necessary to prevent resurrection and prove completion.
- Remove or invalidate embeddings, keyword indexes, topic projections, entity links, summaries, caches, and cross-galaxy derived material.
- Do not recreate the forgotten memory from the same retained source unless the user explicitly restores or reintroduces it under an allowed policy.
- Return a clear completion or failure result; partial cascades must remain visible to administrative repair until complete.

15. Audit, Privacy, and Security

15.1 Audit events

- Creation and source attachment.
- Lifecycle transition and reason.
- User confirmation, correction, rejection, and forgetting.
- Cross-galaxy reads, writes, contributions, authorization grants, revocations, and expired access attempts.
- Export, migration, bulk reindex, and deletion cascade status.

15.2 Security requirements

- Every query must enforce owner scope before ranking or returning records.
- Partition authorization must be evaluated at request time and must not rely solely on previously cached results.
- Source excerpts must be minimized and protected as user data.
- Production canonical data must use the durable database architecture. Repository files may be fixtures, exports, staging, or verified read-only legacy sources only.
- No raw personal exports, uploaded documents, runtime memory, or user-owned canonical records may be committed to Git.
- Backups and exports inherit ownership, retention, and deletion requirements.

16. Existing Object Memory: Preserve, Change, Retire

16.1 Preserve and evolve

Existing capability	Required evolution
Typed objects and properties	Move to the shared object framework with canonical domain and Memory type enforcement.
Source references and evidence excerpts	Move into the source ledger with owner, galaxy, source type, stable source ID, and temporal fields.
Object relationships	Keep as first-class records; add lifecycle validity and authorization-aware traversal.
Confidence	Retain as evidence-proportional extraction confidence; never use it as lifecycle authority.
Status and promotion fields	Replace or map to the locked Memory lifecycle and domain-routing process.
Deterministic extraction	Keep as one reproducible candidate method behind domain classification and Memory settings.
Selective retrieval	Replace keyword-only ranking with multi-index retrieval and mandatory lifecycle/authorization gates.
Backups and reparse history	Retain as migration/audit support, not as competing canonical stores.

16.2 Route existing object types

Current Object Memory type	Target authority
user_profile	Identity
preference	Settings when explicit; Glow when inferred adaptation
project, system, feature, integration, repository, rule, constraint	Primarily Knowledge or Desk / Projects according to canonical ownership
task, open_question	Desk / Projects
decision	Memory when it records a choice; link to affected Project or Knowledge objects
event	Memory when it records an occurrence; system telemetry remains History / ZETA Integrity
memory_conflict	Memory review when personal memories conflict; Knowledge Curation when claims conflict

16.3 Retire as canonical authority

- The undifferentiated user Object Memory graph.
- The claim that all extracted project, system, rule, preference, task, and profile objects are Memory.
- Unscoped system graph fallback as a destination for personal memory.
- Keyword-only retrieval that can return archived, conflicting, under-review, superseded, or unauthorized records.
- Any path that invents an owner or treats a channel identifier as a unified Identity without verification.

17. Migration Requirements

17.1 Migration sequence

Phase	Required result
Inventory	Identify every current memory store, object graph, file fallback, database table, reparse artifact, cache, and retrieval path.
Freeze authority	Declare the new canonical boundaries before migrating records so old writes cannot create new mixed-domain truth.
Classify	Assign each existing object to Identity, Memory, Knowledge, Settings, Glow, Desk / Projects, Files, History, or discard/review.
Resolve ownership	Require a verified user owner. Quarantine rather than guess when ownership is absent or ambiguous.
Assign galaxy	Use verified source context and project ownership. Quarantine ambiguous items for review rather than inferring from a label alone.
Normalize	Create canonical Memory records, source ledger entries, relationships, versions, and lifecycle states.
Reconcile	Merge true duplicates, preserve conflicts, establish corrections and supersessions, and prevent lost provenance.
Index	Build fresh keyword, semantic, topic, entity, temporal, and relationship indexes from canonical records.
Verify	Run ownership, partition, lifecycle, retrieval, toggle, correction, and deletion acceptance tests.
Cut over	Route all reads and writes through the ZCOS Memory Engine; retain old stores only as approved read-only migration evidence until decommissioned.

17.2 Legacy archive rule

The existing zed-memory archive remains historical evidence, not automatic current truth and not shared system memory. This specification does not authorize its deletion or automatic migration. Any future migration requires verified ownership, reconciliation, provenance preservation, and an explicit approved migration run.

18. Required Service Contracts

Exact route names and storage technology are implementation decisions for the build specification. Regardless of interface shape, the engine must expose equivalent authenticated operations for:

- Create a direct confirmed memory.
- Submit or evaluate an inferred candidate.
- List and retrieve memories by You, Topic, Galaxy, entity, and date filters.
- Retrieve ranked context for an authorized response.
- View a memory with source and version history.
- Confirm, reject, correct, supersede, and forget a memory.
- Read and update the Memory toggle and Summary through Settings.
- Execute and verify deletion cascades and index reconciliation.
- Run authorized All Memory queries without collapsing partition identity.
- Record and inspect administrative audit events.

19. Acceptance Criteria

19.1 Ownership and isolation

- User A cannot create, list, retrieve, correct, export, or forget User B's Memory.
- Missing or fallback user IDs fail clearly and create no record.
- An external channel without a verified unified Identity cannot access long-term Memory.

19.2 Partition behavior

- A memory created in ZYNC is stored with ZYNC origin and is available to ZYNC by default.
- ZAR cannot retrieve that ZYNC memory without the required Admin Access authorization.
- Authorized access preserves ZYNC origin in results and audit events.
- Revocation prevents subsequent access and invalidates derived cross-galaxy context.

19.3 Toggle behavior

- When Memory is off, conversation turns create no long-term candidates or records and retrieve no existing memory.
- Existing records remain preserved while off.

- Re-enabling restores eligible access but does not backfill disabled-period conversations.

19.4 Lifecycle and truth

- Proposed records never appear as confirmed user truth.
- A corrected record is used in place of its prior version.
- A superseded record is excluded from ordinary retrieval.
- Conflicting material remains distinguishable and triggers review or inquiry when material.

19.5 Forgetting

- A forgotten memory cannot be found through semantic, keyword, topic, entity, date, relationship, summary, cache, or cross-galaxy retrieval.
- The deletion cascade is idempotent and reports partial failure until every required derivative is invalidated.
- The same source does not silently recreate the forgotten memory.

19.6 Domain routing

- Profile information is not stored as canonical Memory.
- Tasks and open questions are routed to Desk / Projects.
- Facts and system specifications are routed to Knowledge.
- Conversations remain source/history unless a qualifying memory is separately retained.

19.7 Retrieval quality

- Retrieval returns only authorized, eligible, objective-relevant records.
- Directly confirmed corrections outrank older high-confidence extractions.
- Vector index loss can be repaired from canonical records without losing Memory.

20. Implementation Guardrails

Do not build: a second memory graph, a new UI design, automatic disabled-period backfill, a shared personal-memory folder, keyword-only truth retrieval, silent conflict overwrites, or any fallback-owner behavior.

- Preserve the approved simple Settings and Memory interfaces. Backend depth must not increase user-facing complexity.
- Integrate reusable Object Memory components before creating replacements.
- Make the minimum safe migration changes required to establish one authority.
- Do not delete legacy archives or working logic without a separate verified migration and removal plan.
- Treat this specification as the acceptance authority for the Memory Engine build.

Appendix A. Source Authority

Source	Role in this specification
ZCOS Architecture Foundation, August 7, 2026	Primary authority for locked Memory architecture, partitions, lifecycle, UI, domain boundaries, and migration direction.
ZAR Specification, August 4, 2026	Migration evidence for current repository behavior, existing Object Memory, ownership rules, legacy stores, and known exceptions.
ZAR + USER Interaction Guidelines and Behavior	Governs evidence, confluence, context, state tracking, confidence, system integrity, and minimal-change implementation behavior.
ZedAI repository audit	Verifies the existing object types, source references, relationships, graph persistence, keyword retrieval, database memory classes, and fallback safeguards described in the migration sections.